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Neoadjuvant immune checkpoint blockade: A window of opportunity to advance cancer immunotherapy

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SUMMARY

Among new treatment approaches for patients with cancer, few have accelerated as quickly as neoadjuvant immune checkpoint blockade (ICB). Neoadjuvant cancer therapy is administered before curative-intent surgery in treatment-naïve patients. Conventional neoadjuvant chemotherapy and radiotherapy are primarily intended to reduce tumor size, improving surgical resectability. However, recent scientific evidence outlined here suggests that neoadjuvant immunotherapy can expand and transcriptionally modify tumor-specific T cell clones to enhance both intratumoral and systemic anti-tumor immunity. It further offers a unique “window of opportunity” to explore mechanisms and identify novel biomarkers of ICB response and resistance, opening possibilities for refining long-term clinical outcome predictions and developing new, more highly effective ICB combination therapies. Here, we examine advances in clinical and scientific knowledge gleaned from studies in select cancers and describe emerging key principles relevant to neoadjuvant ICB across many cancer types.

INTRODUCTION

Neoadjuvant immune checkpoint blockade (ICB) is being broadly tested in many cancer types, with two FDA approvals within the past two years. Neoadjuvant or “window of opportunity” cancer therapy is administered prior to surgery in treatment-naïve patients with intent-to-cure. Beyond its potential to enhance surgical resectability of the primary tumor, neoadjuvant immunotherapy further seeks to utilize the primary tumor as a source of antigens recognized by the immune system, inducing or enhancing systemic anti-tumor immunity to target and eliminate distant micrometastases that would otherwise become the

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nidus of post-surgical relapse.[1] While neoadjuvant immunotherapies such as interleukin-2, cancer vaccines, and anti-CTLA-4 have been tested in small pilot trials over the past decades, it was the advent of PD-1 pathway blockade, alone or in combination treatment regimens that accelerated the development of this strategy (Figure 1). As evidence of the accelerating activity in this space, only three years elapsed between the first published report of neoadjuvant anti-PD-1 monotherapy in non-small-cell lung cancer (NSCLC)[2] and the first US Food and Drug Administration (FDA) approval for a neoadjuvant anti-PD-1 and chemotherapy combination in triple negative breast cancer (TNBC).[3]

The most rapid clinical endpoint for neoadjuvant therapy is pathologic response in the surgical specimen. With certain neoadjuvant ICB regimens in select tumor types, a pathologic complete response (pCR), i.e., no histologic evidence of residual live tumor cells, is achievable in a substantial proportion of patients. It follows that, if neoadjuvant ICB results in enhanced systemic activation of tumor-specific T cells and pathologic response in the resected tissues reflects systemic immune reactivity against peripheral micrometastatic disease, then one might expect to see a correlation between pathologic response and survival outcomes. Indeed, improvements in both pCR and event-free survival (EFS, absence of disease progression or death after treatment initiation) were observed in the first two FDA approval studies in TNBC[3] and NSCLC,[4] possibly supporting the hypothesized immunologic mechanisms underpinning neoadjuvant ICB and validating pathologic response as an early surrogate endpoint for long-term clinical outcomes. These findings also suggest that limited degrees of pathologic response may warrant additional adjuvant (post-surgical) therapy(ies).

In addition to accumulating evidence for clinical benefit from neoadjuvant ICB, this treatment approach has been a goldmine for conducting detailed correlative immunologic analyses of ICB response and resistance mechanisms. For the first time, large amounts of viable tumor tissue (and sometimes tumor draining lymph nodes and surrounding normal tissues) are available for comparative analyses of pathologic responders vs. non-responders. In particular, high- and ultra-high-dimensional platforms for proteomics, transcriptomics, epigenetics, and B cell receptor (BCR) and T cell receptor (TCR) repertoire analysis allow single-cell analysis of every cell type in the treated tumor. These platforms include high-dimensional flow cytometry and single-cell RNA-seq, ATAC-seq, and TCR-seq. In addition to T cells, B cells, and myeloid cells, stromal cells are also being profiled extensively, generating publicly available databases containing terabytes of information. Finally, high-dimensional spatial proteomic and transcriptomic analyses of intact tumor sections provide additional geographic information. Deconvoluting and mining these rich datasets requires sophisticated computational analysis platforms. Beyond basic discoveries, these scientific endeavors are predicted to guide the continuous improvement of neoadjuvant immunotherapy combinations for the benefit of patients.

Neoadjuvant ICB is a diverse and robust arena of clinical research, with hundreds of ongoing or planned clinical trials in dozens of cancer types. This article will focus on clinical developments in a select group of cancers–NSCLC, TNBC, and skin and gastrointestinal malignancies–in which seminal advances have occurred with broad implications for effectively bringing this treatment approach to other cancer types. We will

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also explore the application of scientific multi-omics investigations within the neoadjuvant treatment framework, to better understand the immunologic mechanisms driving ICB response and resistance.

LUNG CANCER

The treatment landscape for advanced non-small-cell lung cancer (NSCLC) has changed dramatically over the past 10 years, with >25 new drugs, predominantly targeted therapies and ICB, approved since 2010.[5] However, until the 2020 approval of adjuvant osimertinib for resected EGFR-mutant NSCLC, there had been no advances in the systemic treatment of surgically resectable lung cancer for almost 15 years.[6] Outcomes after surgery for NSCLC are among the poorest of all common cancers, with only 50% of patients expected to survive five years after ostensibly curative surgery for stage II (primary tumor ≥4 cm and/or involving the closest lymph nodes) or IIIA (large tumors and/or involving more distant lymph nodes) disease.[7] Adjuvant cisplatin-based chemotherapy confers an overall survival (OS) advantage of ~5% at five years vs. surgery alone; however, this modest benefit has limited routine clinical use.[8] PD-1 pathway-blocking antibodies, either as monotherapy or combined with chemotherapy and/or with a different immune checkpoint blocker, anti-CTLA-4, have become standard-of-care for the majority of advanced unresectable NSCLCs that lack targetable driver oncogenes.[5] Additionally, anti-PD-L1 (durvalumab) therapy after definitive chemoradiation for unresectable stage III disease, clinically termed “consolidation therapy”, improves long-term survival.[5,9] Five-year follow-up from multiple phase 3 trials suggests that approximately 10% of patients with advanced NSCLC experience long-term progression-free survival (PFS), or effective cure, after initial systemic therapy incorporating anti-PD-(L)1.[10–13]

For resectable lung cancer, neoadjuvant chemotherapy confers a similarly limited benefit compared to adjuvant chemotherapy but has potential advantages in terms of tolerability, dose intensity, and *in-vivo* assessment of response through pathologic evaluation of the surgical specimen.[14,15] The first reported clinical trial of neoadjuvant PD-1 pathway blockade in any cancer typewas for resectableNSCLC. Twenty-one patients received nivolumab for ~4 weeks prior to surgery for clinical stage IB-IIIA NSCLC; adjuvant therapy was not mandated.[2] Treatment was well-tolerated, and all but one patient underwent complete surgical resection. Despite the short course of therapy, 45% of patients undergoing resection had a primary tumor major pathologic response (MPR, ≤10% residual viable tumor in the resection specimen[1]). In-depth translational analyses showed rapid expansion of T cell clones shared between the tumor and periphery, including mutation-associated neoantigen (MANA)-specific clones in a patient with pathologic complete response (pCR).[2,16] Indeed, the rapid expansion and contraction of these clones over 4 weeks in the blood was reminiscent of acute responses to viral infection. These findings were consistent with the hypothesis that neoadjuvant ICB may prime systemic anti-tumor immunity while the tumor is still in place. Notably, substantial pathologic responses were observed in several patients despite minimal radiographic response or even apparent tumor growth on pre-operative tumor imaging. This revealed the important theme reproduced in subsequent neoadjuvant ICB studies in other cancer types that ICB-induced regression in tumor tissues can outpace

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radiographic regression. Five-year follow-up from this small trial suggested encouraging outcomes with no active cancers or cancer-related deaths among patients with an MPR.[17]

Among many subsequently reported phase 2 trials, several key studies explored neoadjuvant anti-PD-1 combined with either chemotherapy or anti-CTLA-4. In the NADIM and NADIM-2 trials, patients with resectable stage III NSCLC received standard neoadjuvant carboplatin-paclitaxel chemotherapy with or without nivolumab, followed byoptional adjuvant nivolumab for 6–12 months.[18,19] The single-arm trial, NADIM, enrolled 46 patients and demonstrated a remarkable pCR rate of 57% in the intent-to-treat (ITT) population. Overall survival at 36 months was 82%. The NADIM-2 trial randomized patients to receive 9 weeks of neoadjuvant chemotherapy with or without nivolumab, followed by 6 months of adjuvant nivolumab. The primary pCR endpoint was improved from 7% with chemo alone to 37% with chemo-immunotherapy, while the key secondary OS endpoint was improved from 63% to 85% at 24 months (hazard ratio 0.40, 95% CI 0.17–0.93; $p = 0.034$). Meanwhile, two trials demonstrated potentially enhanced efficacy when anti-CTLA-4 was combined with neoadjuvant anti-PD-1, reporting pCR rates of 33–38% (vs. ~10% with neoadjuvant nivolumab alone), albeit with potential for increased toxicity.[20,21] More recently, the addition of neoadjuvant nivolumab and ipilimumab (anti-CTLA-4) to chemotherapy was reported from the NEOSTAR platform trial.[22] When ipilimumab was added to neoadjuvant nivolumab plus chemotherapy in patients without tumor EGFR or ALK alterations, MPR rates increased to 63% (from 41% with neoadjuvant nivolumab plus chemotherapy), and effector memory $CD8^+$ T, B, and myeloid cells and markers of tertiary lymphoid structures (TLS) were also increased.

Building on this experience, the CheckMate 816 randomized phase 3 trial enrolled 358 patients with clinical stage IB-IIIA NSCLC who received neoadjuvant chemotherapy ± nivolumab for ~12 weeks prior to definitive surgical resection.[4] No adjuvant therapy was required. The study's co-primary endpoints were pCR and EFS. More patients in the nivolumab plus chemotherapy arm completed the planned course of neoadjuvant therapy (94% vs. 85%) and more underwent definitive surgical resection (83% vs. 75%). There was no increase in treatment-related toxicity or surgical complications from adding nivolumab to neoadjuvant chemotherapy, and patients required less extensive surgeries after chemo-immunotherapy. The rate of pCR in the ITT population was significantly increased, from 2.2% with chemotherapy alone to 24% with chemo-immunotherapy (odds ratio 13.94, 99% CI, 3.49–55.75; $p < 0.0001$). The key registrational endpoint of EFS was significantly improved by almost one year with the addition of nivolumab to neoadjuvant chemotherapy (from median 20.8 months to 31.6 months, HR 0.63, 97.38% CI 0.43–0.91; $p = 0.005$). This supported the approval of neoadjuvant nivolumab plus chemotherapy by the FDA in March 2022 and subsequent approvals outside the US. While there was a trend toward improved EFS with neoadjuvant chemo-immunotherapy in patients with higher levels of tumor PD-L1 expression, the study was underpowered for subgroup comparisons, and approvals to date have been agnostic to PD-L1 expression. Tumor mutational burden (TMB) did not predict benefit from neoadjuvant chemo-immunotherapy in this trial.

Notably, in CheckMate 816 patients who received neoadjuvant chemo-immunotherapy and experienced a tumor pCR had a 2-year EFS of >90%, compared to ~50% among patients

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without a pCR (HR 0.13, 95% CI 0.05–0.37). This highlighted the potential use of pCR as an early surrogate of survival benefit from neoadjuvant therapy. The study also examined the dynamics of circulating tumor DNA (ctDNA) in the preoperative period in a subset of 23% of patients enrolled, using a tumor-informed assay. Those patients who eliminated ctDNA before surgery had a trend toward improved EFS (HR 0.60, 0.20–1.82).

Preliminary results from the AEGEAN trial were presented recently.[23] This is the first phase 3 trial to evaluate the role of neoadjuvant chemo-immunotherapy followed by adjuvant immunotherapy, as CheckMate 816 did not mandate any post-surgical therapy. Patients enrolled in AEGEAN were randomized to receive 12 weeks of neoadjuvant platinum chemotherapy plus durvalumab or placebo, followed by surgery, and then up to one year of adjuvant durvalumab or placebo. Both primary endpoints for this study, pCR and EFS, were positive at the first interim analysis. Pathologic CR was increased from 4.3% in the control arm to 17.2% in the durvalumab-containing arm. The hazard ratio for EFS at the time of report was 0.68 (0.53–0.88; p = 0.003902) favoring the durvalumab arm. Notably, these outcomes do not appear to be better than the results from CheckMate 816, despite the addition of adjuvant anti-PD-L1 for one year. A caveat is that only ~31% of patients were evaluable for EFS in the AEGEAN trial at the time of report.

The approval of neoadjuvant nivolumab plus chemotherapy and new results from trials of adjuvant anti-PD-(L)1 have led to resurgent interest in neoadjuvant and perioperative (neoadjuvant plus adjuvant) clinical trials in lung cancer (Table 1). Several more phase 3 trials of neoadjuvant chemo-immunotherapy followed by adjuvant immunotherapy for one year are expected to report in 2024, and the risk:benefit balance from incorporating adjuvant immunotherapy will continue to be a key topic of debate.

TRIPLE-NEGATIVE BREAST CANCER (TNBC)

TNBC (negative for estrogen receptor, progesterone receptor, and HER2) and HER2^{+} breast cancers are aggressive tumors with poor clinical outcomes, leading to the development of standard-of-care neoadjuvant chemotherapy-based regimens for locally advanced disease. This approach sets a precedent for developing neoadjuvant ICB. As with other cancers, neoadjuvant breast cancer therapy has potential advantages over up-front surgery, including downsizing the tumor (potentially avoiding mastectomy and improving cosmetic outcomes), treating micrometastatic disease earlier with systemic therapy, and assessing tumor response on treatment so that alternative drugs may be substituted when initial therapy is ineffective. It also provides prognostic information about long-term clinical outcomes–EFS, OS, and disease-free survival (DFS)–as reflected by the extent of pathologic response present at surgery, where pCR correlates with improved long-term clinical outcomes. The development of immunotherapy in breast cancer has focused on TNBC due to its limited therapeutic targets, more frequent evidence of immune activity, and persistent unmet clinical need. Neoadjuvant ICB strategies are thus a natural fit for early TNBC. After initial clinical trials of anti-PD-(L)1 monotherapy in advanced TNBC showed low response rates (~5–25% depending on prior therapy(ies) and tumor PD-L1 expression),[24] clinical development rapidly shifted to chemo-immunotherapy strategies in both the advanced and neoadjuvant settings.

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The first FDA approval for neoadjuvant ICB as a component of standard treatment for any cancer type was in early stage TNBC.[25] This approval was based on KEYNOTE 522, a randomized phase 3 clinical trial that enrolled 1,174 treatment-naïve patients with stage II-III TNBC.[3,26] The co-primary endpoints were pCR and EFS. Patients were randomized to receive pembrolizumab (anti-PD-1) vs. placebo with paclitaxel and carboplatin for ~12 weeks, then pembrolizumab vs. placebo with anthracycline-based chemotherapy for an additional ~12 weeks, followed by surgery with curative intent. Postoperatively, patients received pembrolizumab or placebo monotherapy for ~27 weeks, for a total treatment period of ~1 year. The pCR rates for neoadjuvant pembrolizumab vs. placebo with chemotherapy were 63% (95% CI 59.6–66.4) vs. 56% (95% CI 50.6–60.6). At a median follow-up of 39.1 months, 36-month EFS rates for pembrolizumab or placebo with chemotherapy were 84.5% and 76.8%, respectively (HR 0.63, 95% CI 0.48–0.82; $p < 0.001$). In contrast to ICB for metastatic TNBC, where PD-L1 expression is required for clinical benefit, in KEYNOTE 522 clinical benefit was observed regardless of PD-L1 expression. Thus, there is currently no predictive biomarker used for selecting TNBC patients to receive neoadjuvant ICB. Based on these data, neoadjuvant pembrolizumab with chemotherapy followed by adjuvant pembrolizumab monotherapy for the treatment of high-risk, early stage TNBC received full FDA approval on July 26, 2021. KEYNOTE 522 highlights multiple opportunities for further investigation, as discussed in the following text.

Precedent for pCR as a surrogate of clinical benefit for regulatory approval of neoadjuvant therapy

An FDA-sponsored meta-analysis of 12 randomized clinical trials in high-risk breast cancer evaluating chemotherapy-based regimens given either before or after surgery demonstrated a strong association between pCR and OS at the patient level (where the pCR surrogate endpoint and OS are strongly associated) but not at the trial level (where the impact of treatment effect on the clinical endpoint is effectively captured (or not) by the surrogate endpoint).[27,28] Although regulatory authorities have supported pCR as a surrogate for long-term clinical benefit (i.e., EFS and OS) in granting accelerated approval in breast cancer, it is notable that before KEYNOTE 522, only the HER2-directed antibody pertuzumab received accelerated approval (in 2013) in the neoadjuvant setting. This was based on the NEOSPHERE trial, which showed a marked improvement in pCR with the addition of pertuzumab to neoadjuvant trastuzumab plus chemotherapy vs. control, with increases in the breast and total (breast plus axilla) pCR rates of 16.8% and 17.8%, respectively.[29] Five-year PFS and DFS rates were in line with the pCR improvement, but the trial was not powered to definitively evaluate these endpoints, and the association of pCR with long-term clinical benefit remains uncertain. This uncertainty was reflected in the FDA approval process for neoadjuvant pembrolizumab with chemotherapy in early TNBC, where the FDA did not grant accelerated approval based solely on significant improvement in pCR,[30] waiting instead for data demonstrating a significant EFS benefit in addition to the pCR improvement before granting full approval.

Extent of pathologic response as a clinical biomarker guiding further management

Beyond being a prognostic marker or surrogate endpoint for clinical benefit of a particular neoadjuvant regimen, pathologic response would be valuable if it could guide

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post-resection therapy decisions. Most straightforwardly, the extent of pathologic response observed at surgery provides an opportunity for treatment escalation or de-escalation, as previously evaluated with conventional treatments for TNBC and HER2$^+$ breast cancer. With neoadjuvant ICB for TNBC, the adjuvant phase of immunotherapy may not be needed for patients who have a pCR. In KEYNOTE 522, patients who achieved a pCR had 4-year EFS rates of 93–94% regardless of whether they had received pembrolizumab with their neoadjuvant chemotherapy or not. Furthermore, the GeparNuevo trial demonstrated that patients who experienced a pCR with neoadjuvant durvalumab plus chemotherapy without an adjuvant phase had a 3-year OS of 100%.[31] Notably, in both studies, patients who received immunotherapy with neoadjuvant chemotherapy and who had residual disease at surgery had significantly better clinical outcomes than those on the placebo arm. Trials that formally evaluate continuation of adjuvant immunotherapy vs. observation in patients who achieve a pCR with neoadjuvant chemo-immunotherapy are warranted. Patients with residual disease after standard neoadjuvant chemotherapy receive adjuvant capecitabine or, in the setting of a germline BRCA mutation, adjuvant olaparib (PARP inhibitor). Many physicians also follow this paradigm when pembrolizumab is part of the regimen, though there are no specific data to validate this decision-making paradigm.

How much and what kind of neoadjuvant chemotherapy are needed for combinations with ICB?

The KEYNOTE 522 regimen utilized 6 months of neoadjuvant chemotherapy with 4 different cytotoxic drugs in addition to pembrolizumab. Importantly, the benefit of including carboplatin with standard neoadjuvant chemotherapy, regardless of whether immunotherapy is also given, remains unclear. Several trials have demonstrated an improvement in pCR with the addition of anti-PD-(L)1 (atezolizumab,[32] pembrolizumab,[33] or durvalumab[31]) to standard neoadjuvant chemotherapy without carboplatin. The NeoTRIPaPDL1 Michelangelo trial tested 6 months of neoadjuvant atezolizumab with nab-paclitaxel and carboplatin vs. chemotherapy alone, followed by surgery and then adjuvant anthracycline-based chemotherapy in 280 patients with early TNBC.[34] There was no difference in the pCR rate between the arms, regardless of tumor PD-L1 expression, suggesting that including an anthracycline in the neoadjuvant therapy phase could be important. In contrast, the single arm, phase 2 NeoPACT trial tested ~4 months of neoadjuvant carboplatin and docetaxel with pembrolizumab, with a pCR rate of 60% and a 2-year EFS rate of 88% in the absence of adjuvant pembrolizumab.[35] Patients with tumors that contained stromal tumor-infiltrating lymphocytes ≥30% had higher pCR rates,[36] suggesting that accurate selection of patients most likely to benefit from immunotherapy may be critical for de-escalating chemotherapy when immunotherapy is given in early disease.

I-SPY2: A neoadjuvant platform trial for predicting clinical benefit and exploring biomarkers

I-SPY2 is a highly innovative, adaptively randomized phase 2 neoadjuvant platform trial that evaluates multiple investigational drugs concurrently against a fixed control arm comprised of standard anthracycline- and taxane-based chemotherapy for 24 weeks.[37] The primary endpoint is the total pCR rate. The investigational treatment graduates for efficacy if the predefined threshold for 85% probability of success in a hypothetical, subtype-specific

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300-patient phase 3 trial is met; it may be discontinued for lack of efficacy, futility, or toxicity. The first ICB arm evaluated was 12 weeks of pembrolizumab concurrent with paclitaxel, followed by 12 weeks of standard neoadjuvant anthracycline-based chemotherapy alone.[38] Pembrolizumab graduated for efficacy after 69 patients (40 with hormone receptor $(HR)^+$ breast cancer and 29 with TNBC) were assigned to pembrolizumab plus paclitaxel relative to 201 control patients who received paclitaxel alone. For HR^+ breast cancer, the estimated pCR rates were 30% vs. 13%, and for TNBC they were 60% vs. 22%. Another arm investigating 12 weeks of pembrolizumab with paclitaxel followed by 12 weeks of pembrolizumab alone was discontinued for lack of efficacy. I-SPY2 also evaluated olaparib with durvalumab and paclitaxel followed by doxorubicin plus cyclophosphamide, which also graduated for both HR^+ breast cancer and TNBC.[39] Other novel neoadjuvant ICB regimens under evaluation in I-SPY2 include pembrolizumab with SD-101 (a TLR9 agonist), cemiplimab (anti-PD-1) with paclitaxel, and cemiplimab with REGN376 (anti-LAG-3). A hypothesis-generating biomarker analysis across five distinct I-SPY2 arms of anti-PD-(L)1 alone or combined with olaparib, SD-101, or anti-LAG-3 demonstrated that tumor immune signatures dominated by chemokines/cytokines were consistently associated with pCR across all anti-PD-(L)1-based arms, regardless of tumor HR status. This finding reflected the co-localization of $PD\text{-}L1^+$ tumor cells and $PD\text{-}1^+$ immune cells.[40] Such platform trial designs rapidly evaluating both the clinical activity of promising novel drugs and exploring potential predictive biomarkers are now expanding to other cancer types.

SKIN CANCERS

Melanoma

As one of the most immunogenic of all human cancers, melanoma has long been an early testing ground for immunotherapies including ICB. It has also been among the earliest platforms for testing neoadjuvant anti-PD-1-based therapies. In contrast to neoadjuvant ICB trials in other cancers, trials in melanoma have focused on treating locoregional stage III metastatic disease and oligometastatic stage IV disease, rather than the primary tumor. This is because localized primary stage I-II cutaneous melanomas typically measure a few millimeters or less, are often curable with surgery alone, and generally have been removed before metastases become evident. Since advanced unresectable melanomas have a high objective response rate to anti-PD-1 monotherapy (~40–50% complete and partial responses), it was anticipated that pathologic response rates with neoadjuvant anti-PD-1 would mirror or exceed this. Surprisingly, four clinical trials published in 2018–2019 collectively demonstrated lower than expected rates of MPR with neoadjuvant anti-PD-1 monotherapy (25–30%); pathologic response rates could be enhanced by adding anti-CTLA-4 to the neoadjuvant treatment regimen (reviewed by Topalian et al.[1]), a strategy that resulted in long-term relapse-free survival (RFS, time to relapse or death after surgery) for patients with a pathologic response of at least 50% (“pathologic partial response”, pPR).[41,42] Concurrently with the development of neoadjuvant therapy for resectable stage III-IV melanoma, two different anti-PD-1 drugs (nivolumab and pembrolizumab) were approved by the FDA as adjuvant (post-surgical) treatments for patients with completely resected stage III-IV melanomas based on results from large randomized trials showing a significantly prolonged time to relapse.[43,44] This created a new benchmark for the further

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development of neoadjuvant ICB in melanoma, because it established adjuvant ICB as an effective treatment paradigm for resectable metastatic melanoma. Preclinical and early clinical data suggested that neoadjuvant ICB could be more effective than adjuvant ICB, by mounting a more potent systemic antitumor immunity while the tumor was still in place.[1,45] Melanoma became the first cancer type in which this question was directly tested, in the SWOG S1801 clinical trial (ClinicalTrials.gov NCT03698019). Here, 313 patients with resectable stage III-IV melanoma were randomly assigned to receive anti-PD-1 (pembrolizumab) either before and after surgery, or only after surgery (standard-of-care adjuvant therapy). All patients were treated for a total of approximately one year. Notably, patients who received neoadjuvant/adjuvant therapy experienced significantly longer 2-year EFS, compared to those who received only adjuvant therapy (72% vs. 49%).[46] Another trial in progress, NADINA (NCT04949113), is addressing the same issue in melanoma by comparing neoadjuvant anti-PD-1 (nivolumab) + anti-CTLA-4 (ipilimumab) followed by surgery and standard adjuvant therapy if >10% residual viable tumor (nivolumab, or BRAF/MEK inhibitors for BRAF-mutant melanoma), vs. immediate surgery and standard nivolumab adjuvant therapy. Although neoadjuvant immunotherapy is not currently FDAapproved in melanoma, some oncologists nevertheless consider this to be a new standard-of-care based on the promising results from the SWOG S1801 trial.[47] Indeed, in May 2023, neoadjuvant/adjuvant pembrolizumab for stage III melanoma was endorsed by the Australian Pharmaceutical Benefits Advisory Committee.

Melanoma investigators were among the first to convene formal stakeholder workshops and collaborative international groups dedicated to advancing neoadjuvant therapies including ICB.[48,49] In particular, early consensus recommendations from the International Neoadjuvant Melanoma Consortium (INMC) for aligning the design, conduct, and interpretation of neoadjuvant ICB trials and correlative scientific studies have influenced the field. The INMC has advocated for innovative surgical approaches such as implanting a tumor marker before initiating neoadjuvant ICB (as is standard in breast cancer), in case near-complete tumor regression renders surgical localization difficult; and performing less extensive surgery in patients who experience a substantial tumor regression. Extending this idea even further, Reijers, Blank, and colleagues conducted the non-randomized PRADO trial in which 99 patients with stage III melanoma received nivolumab plus ipilimumab in the neoadjuvant setting for 6 weeks, followed by biopsy of a previously marked representative “index lymph node” to determine pathologic response.[50] Further therapy was tailored to the level of pathologic response in the index lymph node, including observation alone if pCR or MPR, complete lymph node basin resection if pPR, or complete lymph node resection plus standard adjuvant therapy if >50% viable tumor remaining. However, the occurrence of several relapses among patients achieving pCR/MPR or pPR suggests that some patients may have been undertreated, and highlights the need for more informative biomarkers to personalize neoadjuvant/adjuvant treatment strategies.[51]

Neoadjuvant ICB regimens on the horizon for melanoma include novel treatment combinations, for example, anti-PD-1 (nivolumab) plus anti-LAG-3 (relatlimab),[52] or anti-PD-1 (pembrolizumab) plus anti-TIGIT (vibostolimab);[53] targeting rare melanoma subtypes with favorable genetic features, such as desmoplastic melanomas with very high TMB;[54] and treating high-risk localized primary stage IIB/C disease (ClinicalTrials.gov

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NCT03757689), for which adjuvant anti-PD-1 (pembrolizumab) was recently approved by the FDA.

Non-melanoma skin cancers

Neoadjuvant ICB is being tested in non-melanoma skin cancers including Merkel cell carcinoma (MCC), cutaneous squamous cell carcinoma (cSCC), and basal cell carcinoma (BCC). These locally advanced or metastatic non-melanoma skin cancers occur in relatively rare and elderly patient populations, often arising on the head and neck, and may be neglected by patients for years resulting in challenging clinical presentations. These skin cancers tend to have very high somatic TMB, as they are often caused by chronic exposure to ultraviolet light, a carcinogen. High TMB tends to predict sensitivity to ICB, which is indeed the case for MCC and cSCC. In MCC, high rates of tumor regression and rapid regression kinetics after ICB (nivolumab, pembrolizumab, and avelumab) administered in the advanced unresectable treatment setting[55] translated into a high likelihood of pCR after a brief period of neoadjuvant anti-PD-1 (nivolumab) monotherapy.[56] Notably, 47% of 36 patients with surgically resectable MCC experienced a pCR after only ~4 weeks of neoadjuvant nivolumab monotherapy. Patients with pCR had significantly greater RFS 2 years after surgery, compared to those without pCR (88.9% vs. 52.2%). Likewise, high rates of tumor regression after anti-PD-1 monotherapy (cemiplimab) in advanced unresectable cSCC[57] translated into a 51% pCR rate among 79 patients with surgically resectable disease treated for ~12 weeks in the neoadjuvant setting.[58] Unlike the neoadjuvant ICB experience in melanoma that has focused mainly on treating stage III-IV disease, many patients with MCC or cSCC were treated for locally advanced primary tumors, while others had regional lymph node (stage III) or distant metastases (oligometastatic stage IV). In some patients having tumors in cosmetically sensitive locations on the head and neck, potentially disfiguring surgery was avoided for those with a clinicopathologic CR. Finally, there is limited experience to date with neoadjuvant ICB for resectable BCC, which represents an area of clinical interest (e.g., ClinicalTrials.gov NCT04323202).[59]

GASTROINTESTINAL CANCERS

Neoadjuvant ICB is being tested across a wide range of gastrointestinal cancers, with highly variable efficacy depending on tumor type. Biomarker-driven insights explaining this variability are shedding light on general principles of ICB response and resistance.

Mismatch repair deficient (MMRd) cancers

ICBs have remarkable clinical activity in patients with mismatch repair deficiency (MMRd) cancers characterized by high neoantigen load and their use in this subgroup in the perioperative setting is an area of intense clinical investigation. MMRd is identified approximately twice as often in non-metastatic cancers (8%) as in advanced cancers (4%), representing approximately 40,000 stage I-III cancer diagnoses in the United States annually.[60] The enrichment of MMRd in early stage vs. metastatic cancers may reflect effective immune surveillance in this disease subset, suggesting that non-metastatic MMRd tumors may be particularly amenable to ICB. Furthermore, MMRd tumors often do not benefit from standard perioperative chemotherapy regimens, underscoring the need for

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novel therapeutic approaches.[61,62] Recent prospective neoadjuvant immunotherapy trials in MMRd tumors include NICHE-1 (NCT03026140),[63] NICHE-2 (EudraCT 016-002940-1),[64] and PICC (NCT03926338)[65] in MMRd colorectal cancer (CRC); GERCOR/NEONIPIGA (NCT04006262)[66] and INFINITY in MMRd gastroesophageal cancers;[67] a study in locally advanced MMRd rectal cancer;[68] and a tumor-agnostic MMRd prospective cohort.[69] There are also several retrospective cohort studies.[70–72] Although these studies were conducted in different geographies with different ICB regimens and tumor histologies, they collectively show a low rate of progression events and high pathologic response rates across all cohorts, with pCRs in ~60% of patients in the largest studies reported to date.

The pathologic response rates with neoadjuvant immunotherapy in non-metastatic MMRd cancers reported to date are substantially higher than those in the metastatic setting; even within non-metastatic tumors, response rates may correlate with tumor stage. For example, in the INFINTY study the pCR rates were 89% (8/9) in patients with T2-T3 disease vs. 17% (1/6) in T4 disease.[67] This may reflect increasing genetic heterogeneity and immune resistance as these tumors progress, and provides a justification for considering the use of ICB as early as possible in MMRd tumors. Pathologic response rates appear similar in MMRd CRC and MMRd gastric cancers but are potentially lower in other histologies. In the only tumor type-agnostic series to date, 2/2 patients with potentially resectable MMRd pancreatic cancer experienced disease progression with ICB and required salvage surgery.[69] This is similar to the metastatic setting, where pancreatic MMRd tumors appear to be less responsive than other MMRd tumor types.[73] Thus, the clinical benefit of neoadjuvant ICB in non-metastatic MMRd cancers may vary not only by cancer stage but also by tumor histology. As non-metastatic MMRd tumors have distinctive operative risks depending on cancer stage and tumor histology, as well as distinct surveillance challenges, neoadjuvant ICB approaches may eventually need refinement within each tumor type and stage to optimize patient outcomes.

The exceptional pCR rate achieved in resectable MMRd tumors paves the way toward new non-operative organ preservation approaches. In a study by Cercek and colleagues, 100% (12/12) of patients with locally advanced MMRd rectal cancer treated with 6 months of anti-PD-1 (dostarlimab) had complete radiographic and endoscopic responses without evidence of residual tumor on serial biopsies for ≥6 months. Conventional chemoradiotherapy plus surgery, interventions that often cause life-long bowel and bladder dysfunction and infertility, were completely avoided in this cohort.[68] These findings, though in a small cohort, set the stage for neoadjuvant ICB as an effective approach to completely eliminate conventional procedures that may create significant lifelong effects on quality of life even if the cancer is cured. In a separate, real-world cohort of surgically unfit patients with locally advanced or oligometastatic CRC, 13/23 (57%) achieved a complete radiographic response, potentially obviating the need for surgery.[74] Larger multicenter cohorts with longer-term follow-up are needed before an organ-preservation strategy can be declared a new standard-of-care for patients with MMRd, but available data suggest that in many cases immunotherapy alone can be definitive ablative therapy for patients with early stage MMRd CRC.

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Hepatocellular carcinoma (HCC)

In HCC, the feasibility of neoadjuvant ICB was recently demonstrated in three separate single-institution studies.[75–77] These pilot studies demonstrated MPR rates of 20–33% for nivolumab plus cabozantinib (multi-kinase inhibitor), nivolumab with or without ipilimumab, and cemiplimab, with outstanding survival in the subset of pathologic responders. While many neoadjuvant ICB studies do not primarily aim to improve tumor resectability, the small pilot study of neoadjuvant nivolumab plus cabozantinib in HCC was unique in its inclusion of potentially resectable patients outside of traditional resection criteria (such as patients with major blood vessel involvement or multifocal disease) and its use of resection with negative parenchymal margins as a major endpoint. Adjuvant atezolizumab plus bevacizumab (angiogenesis inhibitor) was recently reported to improve RFS in patients with HCC following resection or ablation, and is likely to be adopted as a standard-of-care for patients at high risk for recurrence.[78] However, only ~20% of patients with HCC have a tumor that is resectable by Barcelona Clinic Liver Cancer (BCLC) guidelines.[79] The preliminary success of neoadjuvant immunotherapy in HCC may expand the pool of patients who can ultimately be considered for curative intent paradigms (Figure 2).

Gastroesophageal cancers

In gastroesophageal cancers, where anti-PD-1 therapy has established efficacy in the metastatic and adjuvant settings, neoadjuvant ICB is an area of active clinical research. The randomized phase 2 DANTE study of the FLOT-AIO German Gastric Cancer Group and Swiss SAKK group is comparing standard-of-care perioperative chemotherapy (FLOT) vs. FLOT plus atezolizumab. Patients in this study are stratified by clinical lymph node stage and primary location (gastroesophageal junction vs. stomach) as well as MMR status. Interim results showed a modest improvement in the rate of pathological response in the atezolizumab-containing arm (49% vs. 39%). The benefit of atezolizumab appeared to be more pronounced in the subset of tumors with high PD-L1 expression (CPS > 10%), where pathological response rates were 67% for FLOT plus atezolizumab vs. 39% for FLOT alone.[80] Although these preliminary results require further confirmation, they indicate that tumor PD-L1 expression may be a useful biomarker to select patients with gastroesophageal cancers for neoadjuvant immunotherapy.

Pancreatic ductal adenocarcinoma (PDAC)

Pancreatic ductal adenocarcinoma (PDAC) is a notably immune-resistanttumor in which immunotherapy has thus far failed to conclusively show clinical benefit in any disease setting, and OS remains dismal (5-year survival <9%). In most cancers, neoadjuvant immunotherapy studies have generally been conducted only after demonstration of efficacy in the advanced disease setting. However, in pancreatic cancer, novel immunotherapeutic agents have been deployed in window of opportunity studies long before demonstration of clinical efficacy in the advanced treatment setting,[81] providing rapid pathologic insights about potential efficacy and the mechanisms of response to therapy. For example, the recently initiated PIONEER-Panc phase 2 study uses a Bayesian platform design to evaluate one or more experimental arms simultaneously against a common chemotherapy control

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arm in patients with resectable, borderline resectable, and locally advanced disease.[82] This innovative drug development framework has precedent in the I-SPY2 platform trial in breast cancer described previously. It reflects the critical need for new therapies in PDAC and offers a blueprint for rapid evaluation and evolution of novel therapeutics that can be applied across tumor types.

IMMUNOLOGIC ANALYSIS OF NEOADJUVANT ICB TRIALS: A GOLDMINE FOR REVERSE TRANSLATION

The neoadjuvant ICB setting is uniquely positioned to address fundamental questions surrounding mechanisms of treatment response and resistance, owing to the ability to obtain substantial amounts of viable tissue and blood before and after ICB (Figures 1 and 3), which is not routinely possible with advanced unresectable cancers. Acquisition of these precious biospecimens must be a top priority in the design of every neoadjuvant ICB trial. Coordination among oncologists, surgeons, pathologists, and laboratory scientists is critical to provide a comprehensive blood and tissue collection plan to produce high-quality tissue and cellular samples for accurate analyses, particularly for those requiring viable cells. Even in tumors with partial or complete pathologic regression at the time of surgery that may have limited or no viable tumor cells available for analysis, the remaining tumor bed can still serve as a wealth of important and relevant immunologic information. In this section, we will discuss immunologic investigations that can be performed on biospecimens from neoadjuvant ICB studies to define biomarkers of response, discover new actionable targets and pathways, and advance our basic understanding of how ICB impacts cells systemically and locally in the tumor immune microenvironment (TIME). For these studies, high-dimensional analyses, such as high-parameter flow cytometry, single-cell (sc)RNA-seq and coupled scTCR-seq, and spatial genomics and proteomics are transforming our ability to understand the effects of ICB on the TME. Translational impacts of these studies are providing potential predictive biomarkers to guide post-surgical adjuvant therapy and discover immunoregulatory molecules and cells that can be therapeutically targeted as an adjunct to ICB with currently approved antibodies.

Pathologic-spatial analysis of resected tumors after neoadjuvant ICB

The most fundamental analysis of the TIME in the context of ICB comes from pathologic scoring of percent residual viable tumor in resection specimens. However, a more detailed analysis of immunologic features in post-ICB tissue specimens has the potential to refine predictions of long-term EFS/RFS/OS. Similar to the advanced metastatic setting for most cancer types, two conventional tests to predict ICB response, PD-L1 expression in the tumor and TMB, have thus far been imperfect biomarkers in the setting of neoadjuvant ICB. For neoadjuvant trials in NSCLC, association of these parameters with outcomes has been variable across different studies. In one study, there was a correlation between TMB and percent residual tumor following neoadjuvant anti-PD-1 (nivolumab),[2] although the number of patients for whom TMB estimates could be generated was small (n = 11). TMB from these 11 patients was not associated with improved RFS or OS.[17] Furthermore, tumor PD-L1 expression was not associated with the degree of pathologic responsein this study. Conversely, in a report by Reussetal., lung tumor PD-L1 expression but not

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TMB was predictive of pathologic response to neoadjuvant anti-PD-1 (nivolumab) plus anti-CTLA-4 (ipilimumab), although the number of evaluable patients in this study was also small (n = 6).[21] In a slightly larger study in which resection specimens from 19 patients with locally advanced gastric cancer were evaluable, both PD-L1 and TMB correlated with pathologic response toneoadjuvant anti-PD-1 (camrelizumab) combined with apatinib (VEGFR-2 kinase inhibitor) and chemotherapy.[83]

While pathological response as defined by residual tumor at resection is becoming an established biomarker for long-term clinical benefit, as described above, more sophisticated analyses of tumor sections have the opportunity to further refine clinical decision-making as well as understand mechanisms of response vs. resistance to ICB. Cottrell et al. have proposed immune-related pathologic response criteria (irPRC) to evaluate certain histologic features associated with neoadjuvant ICB treatment.[84] Not only does this method inform about intratumoral immune events that occur during neoadjuvant ICB and are associated with pathologic response, but it reveals a major source for the kinetic discrepancy between pathologic and radiographic response, namely that the regression bed in MPR tumors is occupied and characterized by several TIME remodeling features including neovascularization, cholesterol clefts from tumor cell death, and proliferative fibrosis.[84] Such features may appear as a solid mass on conventional radiographic scans. More detailed spatial studies have found that neoadjuvant ICB has the potential to induce diverse inflammatory infiltrates including $CD8^+$ tumor-infiltrating lymphocytes (TIL), neutrophils, and plasma cells; it can also upregulate certain immune checkpoint pathways, reduce numbers of immunosuppressive M2 macrophages, and generate organized TLS in the tumor bed.[52,63,81,85–91] These effects can be modulated by the type or combination of neoadjuvant ICB. For example, as mentioned previously, the addition of ipilimumab to neoadjuvant chemotherapy plus PD-1 blockade resulted in increased expression of $CD8^+$ T cell-activation markers and attenuated expression of markers associated with immunosuppression in resectable lung cancers treated on the NEOSTAR trial.[22] Some of these parameters may be associated with pathological response vs. resistance and/or long-term clinical outcomes. In a pancreatic cancer neoadjuvant ICB trial comparing a GM-CSF-secreting vaccine with or without anti-PD-1, an increase in degranulating neutrophils in the TIME was associated with inferior RFS.[81] There have also been reports supporting a thus-far under-appreciated role for $CD4^+$ T_{conv} cells in the ICB response. Li et al. found that ICB impacts $CD4^+$ T cell chemotaxis and could promote the development of TLS that become sites of immune regulation,[81] while another study suggests that tumors with MPR have greater $CD4^+$ T cell infiltrates.[92] It has also been suggested that $CD4^+$ T follicular helper cells reactive to *E. coli* and other commensal bacteria are the main targets of PD-1 blockade and facilitate clinical efficacy to neoadjuvant ICB in bladder cancer,[86] although these findings may be unique to this cancer type and their *in vivo* relevance requires further validation.

The role of non-hematopoietic cells has also become increasingly appreciated in the neoadjuvant ICB setting. For instance, in Merkel cell carcinomas regressing after neoadjuvant anti-PD-1 (nivolumab) therapy, a prominent stromal myofibroblast signature was identified using geographic bulk RNA-seq, consistent with tissue remodeling occurring after tumor cell elimination.[93] Although most of the correlative studies discussed here have

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been performed in small patient cohorts, they have inspired follow-up studies in larger cohorts in ongoing trials.

Taking these multi-omic analyses to an even higher level, emerging spatial transcriptomic technologies with unprecedented resolution havethe potential to inform on direct cellular interactions underpinning response and resistance to neoadjuvant ICB, which is requisite for identifying new pathways and molecules for therapeutic targeting. For example, a recent study of neoadjuvant anti-PD-1-treated HCC integrated coupled scRNA-seq/TCR-seq with multiple spatial mRNA (MERFISH, RNAscope) and multiplex immunofluorescence staining to identify TIME features associated with pathologic response.[94] In this study, pathologic response correlated with co-localized "cellular triads" containing PD-1^{hi} CD8^{+} T effector cells, CXCL13^{hi} CD4^{+} T_{conv} cells, and activated DC expressing multiple costimulatory molecules (CD80/86) and high MHC II along with multiple co-inhibitory molecules (PD-L1/L2). These findings are compatible with the hypothesized role for tumor antigen cross-presentation as an important mechanism for generating anti-tumor immunity by neoadjuvant ICB.

Neoantigen-specific T cells

Because ICB efficacy is predicated on its ability to (1) enhance priming of *de novo* anti-tumor T cells, (2) "reinvigorate" dysfunctional (often termed exhausted) tumor-specific T cells, and (3) unleash direct tumor killing, it serves that tumor-specific T cell functionality prior to and in response to neoadjuvant ICB is a key determinant of clinical benefit. While most studies look at T cells broadly without regard for antigen specificity, it is ultimately *tumor-specific* T cells whose numbers and functional states are most relevant to anti-tumor immunity. Detection of these cells is challenging in most cancer types, as their frequency is relatively low even within the tumor, and is orders of magnitude lower in the peripheral blood. Using tetramer isolation, Rizvi et al. first demonstrated that neoantigen-specific T cells could be detected in the peripheral blood of a patient with advanced metastatic NSCLC treated with anti-PD-1.[95] These T cells increased in frequency after PD-1 blockade and were associated with a radiographic response. While this was only a single-patient case study, it laid the foundation for comprehensive immune profiling in patients receiving ICB in the neoadjuvant setting. Indeed, neoantigen-specific T cells were subsequently detected in the peripheral blood of a patient with NSCLC treated with neoadjuvant anti-PD-1,[2] and the peripheral dynamics of these T cell clonotypes were concordant with the clinical course of this patient. In additional patients from the same trial, it was found that T cell clonotypes that occupied the most "space" within the tumor also expanded in the periphery upon treatment, which was associated with a pathologic response.[16] Consistent with this, a clonal expansion score and proliferation of hyperexpanded peripheral blood T cell clones correlated with partial pathologic response in a neoadjuvant ICB study in gastric cancer.[83] Furthermore, in a small randomized study comparing neoadjuvant/adjuvant vs. adjuvant ipilimumab + nivolumab in stage III melanoma, peripheral expansion of tumor-resident T cell clones with neoadjuvant/adjuvant ICB was greater than with adjuvant ICB alone.[45] Of note, these three studies[16,45,83] did not confirm the specificity of the expanded clones but were consistent with the hypothesis that neoadjuvant ICB can induce systemic expansion and activation of tumor-reactive T effector cells that have the potential to seek out and

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destroy distant micrometastases. Ultimately, ongoing studies analyzing the dynamics of validated tumor-specific T cells will provide a critical understanding of how ICB modulates anti-tumor T cell immunity.

Transcriptional T cell profiling and response to neoadjuvant ICB

Recently, the integrated analysis of bulk TCR-seq of antigen-stimulated T cells, coupled with scRNA-seq/TCR-seq, has allowed the transcriptional profiling of individual tumor-specific T cells within tumor, tumor-draining lymph nodes, and peripheral blood of patients treated with neoadjuvant ICB.[96] In the original study of $CD8^+$ T cells from neoadjuvant ICB-treated patients with NSCLC using these techniques, Caushi et al. showed that neoantigen-reactive TIL could be detected at the single-cell level in both MPR and non-MPR tumors, and these cells largely possessed a tissue-resident memory (TRM) program. Even though the frequencies of neoantigen-specific $CD8^+$ T cells were not higher in MPR vs. non-MPR tumors, TIL from non-MPR tumors had higher immune checkpoint expression, lower *IL7R* expression, and consequently a blunted response to IL-7 stimulation, implying a dampened tumor antigen-specific proliferative capacity relative to TIL from MPR tumors.[96] In support of the systemic impact of ICB, this study also found that neoantigen-specific T cells from a patient whose tumor had a pathologic CR expanded transiently in peripheral blood and underwent reprogramming from a memory to an effector-like state.

A study by Luoma et al. applied coupled scRNA-seq/TCR-seq to head and neck squamous cell carcinomas from patients receiving neoadjuvant nivolumab with or without ipilumumab. Similar to Caushi et al., Luoma showed that $CD8^+$ TIL with specificity for tumor-associated antigens (TAA) virtually all resided in a TRM cluster.[97] They analyzed tissues pre- and post-ICB, showing a clonal expansion of TAA-specific TRM cells present pre-therapy as well as a minority of new clones post-ICB that were not detected pre-treatment. MPR tumors had a far greater expansion of pre-existing and *de novo* TAA-specific clones than non-MPR tumors, and these clones also expanded in the blood. Even though TAA-specific TIL retained TRM transcriptional programs post-ICB treatment, they had increased expression of cytotoxic genes as well as certain activation-induced immune checkpoint genes, such as *LAG3*.

Among the multiple cell types in the TIME that can limit anti-tumor effector T cell responses, regulatory T cells (T_{reg}) stand out in murine models, in which genetic depletion of $FOXP3^+$ T_{reg} cells mediates the regression of even aggressive tumors that do not respond to conventional ICB.[98] A recent scRNA-seq analysis of tumor-infiltrating T_{reg} from neoadjuvant anti-PD1 treated NSCLC revealed nine different sub-populations.[99] One population, defined by high levels of *OX40, GITR*, and *LAG3*, possessed high suppressive activity and was more frequent in non-responding tumors. Conversely, a population of T_{reg} with a Th1 signature was associated with response. These unexpected findings suggest that targeting specific sub-populations of T_{reg} will have distinct consequences for anti-tumor immunity. For example, the failure of OX40 and GITR agonist antibodies to exert clinical benefit in cancer patients may be because they are actually stimulating suppressive sub-populations of $TILT_{reg}$ to inhibit anti-tumor T effector responses. Although to date the

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number of patients for whom tumor-reactive TIL has been studied is small, these studies have laid the foundation for experimental and computational methods to interrogate the functional programming of the cells that are directly targeted by ICB: those recognizing the tumor.[1]

Operationalizing correlative science to maximize success

Neoadjuvant ICB has the potential to yield a wealth of biospecimens and, consequently, a wealth of information surrounding the mechanistic underpinnings of response and resistance to immunotherapy. While the majority of in-depth scientific analyses reported thus far have focused on T cell responses, multiple non-T cell elements of the TIME (and tumor-draining lymph nodes), including non-hematopoietic cell types, may impact anti-tumor immunity; these elements must be profiled concordantly with relevant tumor-specific T cells, including with spatial co-localization. With proper planning, organization, and a “team science” approach, many impactful correlative studies can come from a single clinical trial, which will ultimately drive biomarker and therapeutic development to improve patient outcomes.

DISCUSSION

Even though follow-up for most neoadjuvant ICB clinical trials is still relatively short, a number of themes appear to be emerging, particularly from the growing number of trials with a median follow-up ≥2 years. First, for a given tumor type, neoadjuvant anti-PD-(L)1 and anti-PD-(L)1-containing combination regimens that show significant efficacy against advanced unresectable disease also tend to mediate a substantial proportion of pathologic responses when given prior to surgery. That said, it is impossible to directly compare pathologic responses in the neoadjuvant setting to clinical responses according to radiographic criteria typically used to assess tumor regression in unresectable cancers, since radiographic responses after ICB tend to underestimate pathologic responses, as discussed above for NSCLC.[2] In pathologic MPRs (including pCRs), viable tumor is replaced by fibrosis, lymphoid infiltrates, and tertiary lymphoid structures resembling lymph node follicles, which may appear as measurable tumor on radiographic scans.

A second major theme emerging from neoadjuvant ICB is that the depth of pathologic response correlates with the duration of EFS/RFS. In the two successful neoadjuvant ICB plus chemotherapy registration trials in TNBC and NSCLC, there were few relapses among patients achieving a pCR at the time of surgical resection. This suggests that pathologic response is a surrogate for the successful induction of systemic anti-tumor immunity and the elimination of micrometastases that are not visible radiographically at the time of surgery. Taube and colleagues have refined this concept in lung cancer, demonstrating a graded correlation between the proportion of residual viable tumor and EFS, and suggesting opportunities for a refined prognostic grading system.[100] Indeed, some patients with even minor pathologic responses in the primary lung tumor can still experience prolonged EFS, suggesting that primed anti-tumor T cell responses can potentially be more effective against distant micrometastatic disease than against bulky primary tumors. More sophisticated spatial analyses of specific cellular markers and marker combinations, applied to resected tumors post-neoadjuvant therapy, will likely be more revealing of systemic responses than

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the simple assessment of percent residual viable tumor on routine H&E staining. Ultimately, information gleaned from interrogating the post-neoadjuvant tumor specimen may guide the personalized application of post-surgical (i.e., adjuvant) therapies to further enhance patient outcomes. Although many current neoadjuvant trials are designed to continue ICB after surgery, this comes with the risk of toxicity, and there may be patients such as those achieving a pCR who do not require continued therapy. In addition to more sophisticated analyses of resected tumors, improved serum/plasma ctDNA and proteomic technologies to detect minimal residual disease will likely emerge as important biomarkers, alone or in conjunction with pathologic response, to guide post-surgical treatment or observation.

A third emerging theme is that neoadjuvant ICB may not only prime systemic anti-tumor immunity but can also affect surgical resection in a positive way. This has long been the case for standard neoadjuvant chemotherapy in high-risk early breast cancer. For highly ICB-responsive cutaneous cancers commonly affecting the head and neck, such as MCC and advanced cSCC, for which conventional extirpative surgery can be disfiguring, tumor reduction with neoadjuvant ICB can reduce the surgical field and significantly improve cosmetic and functional outcomes. Likewise, in MMRd rectal cancers, neoadjuvant ICB can avoid the need for conventional life-altering surgery; and in primary HCC, many tumors that are borderline resectable at diagnosis with a high risk of post-surgical relapse can be successfully treated with intent-to-cure surgery after neoadjuvant ICB. Indeed, for certain cancers, such as melanoma and MMRd rectal carcinoma, the possibility of avoiding surgery entirely in patients achieving a complete pathologic or clinical response is being explored in clinical trials.

On the scientific side, single-cell analysis of viable resection specimens post-neoadjuvant ICB, particularly with high-dimensional flow cytometry and coupled RNA-seq/TCR-seq to profile validated tumor neoantigen-specific T cell clones, has yielded important insights correlating unique T cell transcriptional programs with immunotherapy responsiveness. A majority of tumor-specific T cells express CD39, CXCL13, and LAG-3 and have transcriptional programs with features of dysfunction (high checkpoint expression and killer inhibitory receptor expression), tissue residence (high *CD103* and the canonical tissue-resident transcription factor *Hobit*) and partial activation (e.g., high MHC class II expression). Comparison of tumors with pathologic response vs. non-response after neoadjuvant ICB shows that $CD8^+$ T cells from responding tumors have lower immune checkpoint expression, higher expression of *IL7R*, and a general program skewed toward effector CTL. In contrast to $CD8^+$ effector cells, T_{reg} may represent a significant immune inhibitory force within the tumor. As described above for NSCLC, T_{reg} from tumors responding to neoadjuvant ICB have a distinct subset composition compared to non-responding tumors. The quantum leap in information gained from high-dimensional analyses, particularly single-cell and spatial, also reveals striking differences in the immune microenvironment in different tumor types. Furthermore, such analyses have the potential to shed light on the sometimes-observed heterogeneous clinical behavior of multiple tumor lesions in individual patients, by permitting lesion-level biomarker analysis.[101]

With an explosion of scientific insights and clinical benefits emerging from neoadjuvant ICB and its combinations with targeted kinase inhibitors, chemotherapies, other immunotherapies

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and experimental agents, this field of translational investigation is still just scratching the surface of possibilities. Hundreds of neoadjuvant ICB trials are now underway–they must be carefully designed to answer the most salient clinical and scientific questions in this new age of cancer immunotherapy.

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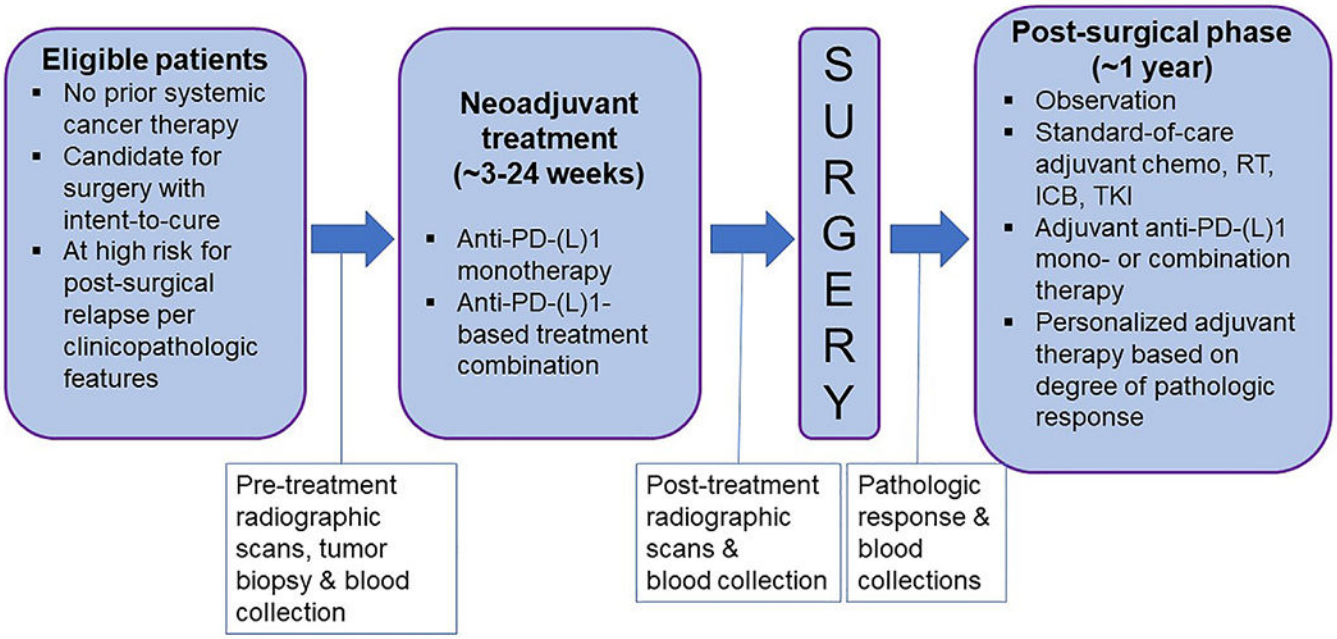

Figure 1. Treatment paradigm for neoadjuvant immune checkpoint blockade

Patients enrolled in clinical trials of neoadjuvant immune checkpoint blockade are typically naive to other systemic cancer therapies and have cancers that are deemed potentially resectable for cure by a surgical expert but with clinicopathologic features associated with a high risk for relapse. Depending on cancer type, the neoadjuvant treatment period may range from ~3 to 24 weeks, followed by a surgical procedure, which can be that which was originally planned, or tailored according to clinical/radiographic evidence of tumor regression. Pathologic response is determined in the surgical specimen. After surgery, patients may enter an observation phase, receive a standard-of-care or experimental adjuvant therapy, or receive an adjuvant therapy tailored to their degree of pathologic response. Blood and tissues collected before and after neoadjuvant ICB are used for multi-omics correlative studies supporting biomarker discovery. Chemo, chemotherapy; ICB, immune checkpoint blockade; RT, radiation therapy; TKI, tyrosine kinase inhibitor.

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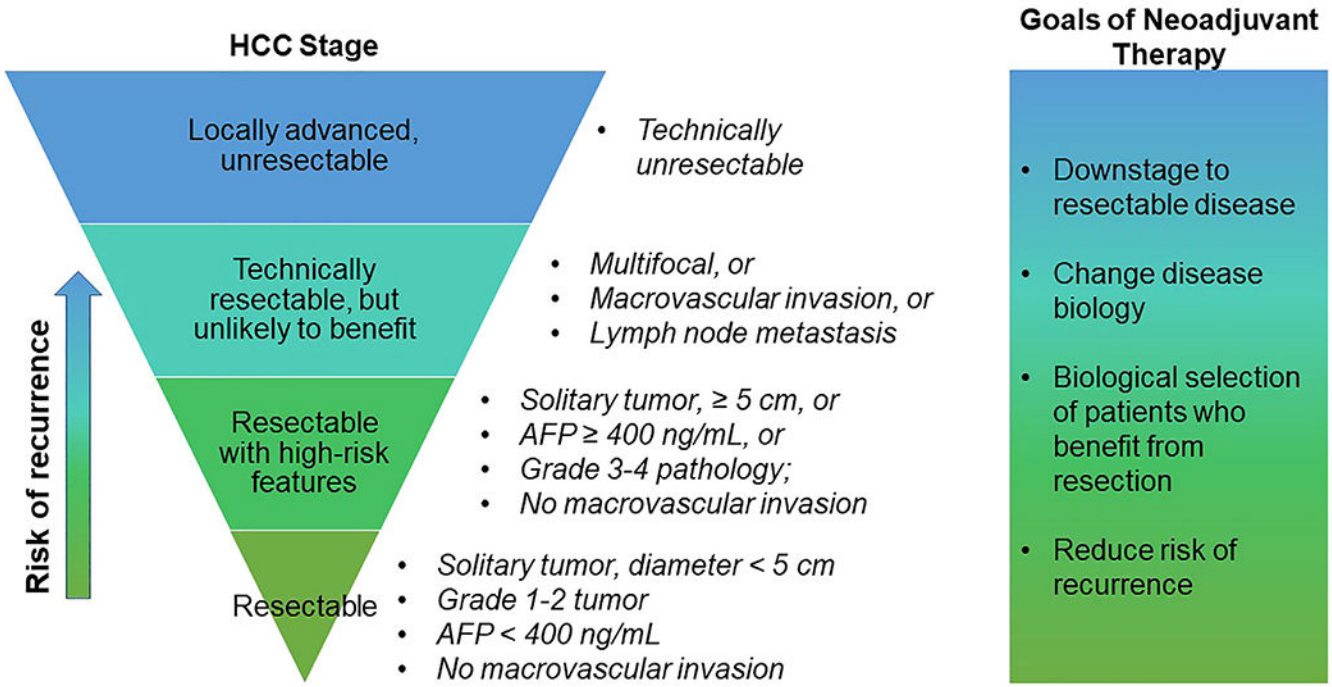

Figure 2. Neoadjuvant therapy downstaging/resectability strategies in hepatocellular cancer

Only a subset (~20%) of HCC is resectable, and in the subset of patients who undergo potentially curative resection, 70% or more will eventually recur. Both neoadjuvant and adjuvant immunotherapy treatment strategies may offer the possibility of reducing the risk of recurrence, however neoadjuvant immunotherapy also offers the possibility of testing disease biology to select patients who may benefit from resection, and expanding the population who may be considered for resection by changing disease biology and downstaging patients whose disease is locally advanced or unresectable. AFP, alpha-fetoprotein.

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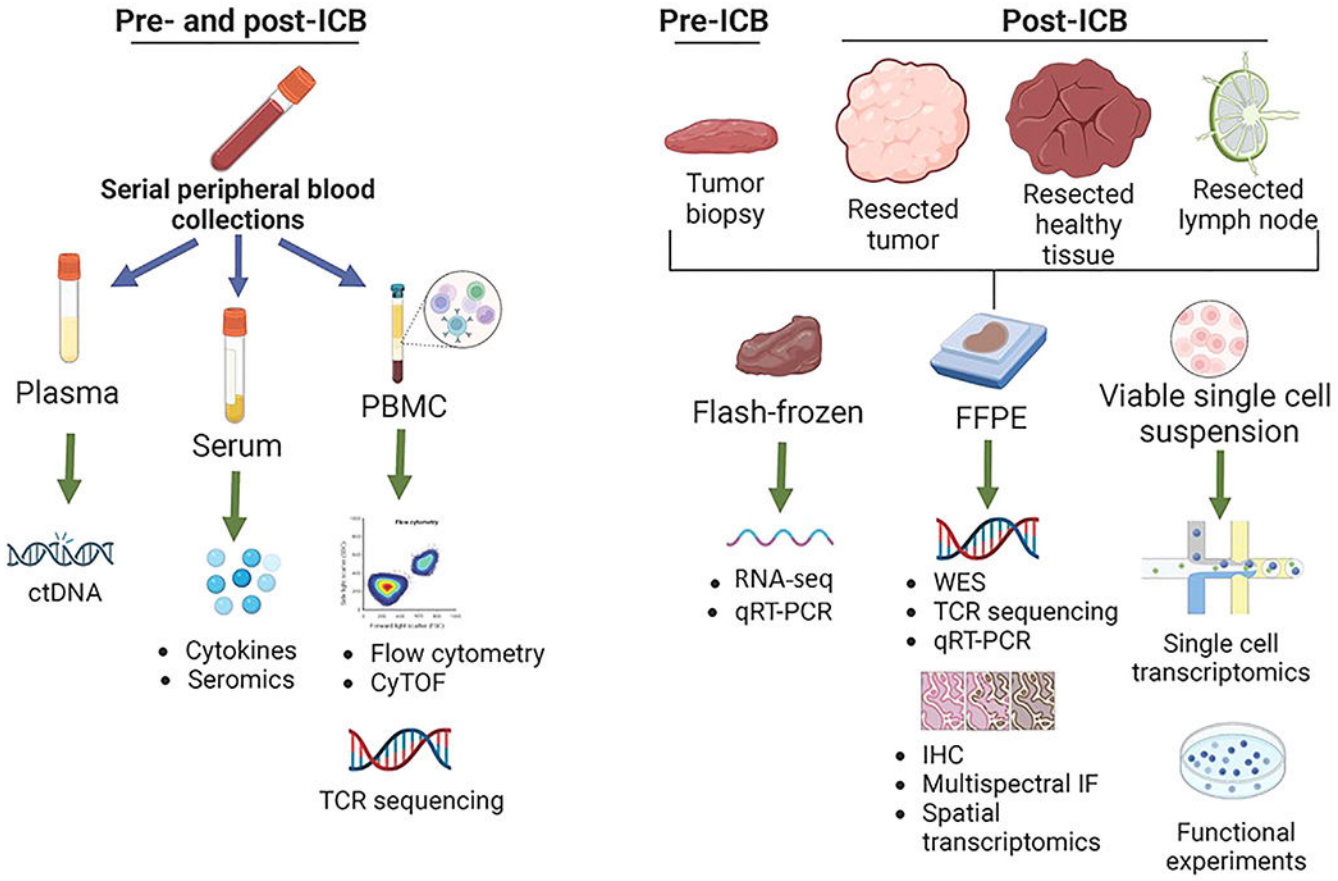

Figure 3. Biospecimen workflow for multi-omics studies integrated with neoadjuvant ICB

Neoadjuvant studies offer a unique opportunity to acquire blood and substantial quantities of viable tissues before, during, and after immune checkpoint blockade (ICB) treatment, which is not routinely achievable with most unresectable cancers. Peripheral blood obtained at serial timepoints can be used to isolate plasma, serum, and peripheral blood mononuclear cells (PBMC). These samples can then be used downstream to analyze circulating tumor DNA (ctDNA), cytokines, and antibodies (seromics), as well as immune cell surface and intracellular protein expression using flow cytometry and cytometry by time of flight (CyTOF), and T cell repertoire using T cell receptor (TCR) sequencing. In addition, pre-treatment tumor biopsies and resected tumor, neighboring healthy tissue, and tumor-draining lymph nodes can be preserved in a variety of ways according to the desired downstream analyses, including flash-frozen, formalin-fixed and paraffin-embedded (FFPE), and viably cryopreserved enzymatically digested single-cell suspensions. These tissue archiving methods will allow for RNA sequencing (RNA-seq), qRT-PCR, whole exome or genome DNA sequencing (WES, WGS), TCR sequencing, immunohistochemistry (IHC), multispectral immunofluorescence (mIF), spatial transcriptomics, functional cellular assays, and single cell transcriptomics, among other possibilities. Figure created with Biorender.com.

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Table 1.

Select phase 3 perioperative ICB trials in non-small-cell lung cancer

ClinicalTrials.gov identifier	Study name (patients enrolled/planned)	NSCLC stage (AJCC edition)	Neoadjuvant chemotherapy backbone[a]	Neoadjuvant ICB	Adjuvant ICB	Primary trial endpoints
NCT02998528	Checkmate 816 (n = 358)	IB-IIIA (7th)	3 cycles[b] of Cis/Carbo + Vin/Peme/Gem/Doce/Pacli	+/− Nivo	No	pCR EFS
NCT03425643	Keynote 671 (n = 786)	IIA-IIIA (8th)	4 cycles of Cis + Peme/Gem	+ Pembro or placebo	Pembro or placebo for one year	EFS OS
NCT03456063	IMPOWER 030 (n = 450)	II-IIIB (8th)	4 cycles of Cis/Carbo + nab-Pacli/Peme/Gem	+/− Atezo	Atezo or best standard care for 1 year	MPR EFS
NCT03800134	AEGEAN (n = 780)	IIA-IIIB (8th)	4 cycles Cis + Gem/Peme or, Carbo + Peme/Pacli	+ Durva or placebo	Durva or placebo for 1 year	pCR EFS
NCT04025879	CheckMate 77T (n = 452)	II-IIIB (8th)	3-4 cycles Cis/Carbo + Peme/Doce or, Carbo + Pacli	+ Nivo or placebo	Nivo or placebo	EFS

AJCC, American Joint Committee on Cancer; Atezo, atezolizumab (anti-PD-L1); Carbo, carboplatin; Cis, cisplatinum; Doce, docetaxel; Durva, durvalumab (anti-PD-L1); EFS, event-free survival; Gem, gemcitabine; ICB, immune checkpoint blocker; MPR, major pathologic response; nab-Pacli, albumin-bound nanoparticle paclitaxel; Nivo, nivolumab (anti-PD-1); OS, overall survival; Pacli, paclitaxel; pCR, pathologic complete response; Pembro, pembrolizumab (anti-PD-1); Peme, pemetrexed; Vin, vinorelbine.

[a]Slashes indicate physician's choice of one chemotherapy drug in the group.

[b]One cycle is 3 weeks in duration.